

# A Preliminary Evaluation of the Efficacy and Safety of Cogent db (an Ayurvedic Drug) in the Glycemic Control of Patients with Type 2-Diabetes

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## ABSTRACT

**Objectives and study design:** A nonrandomized, non-placebo-controlled clinical trial to evaluate the efficacy of Cogent db (an herbal preparation; Cybele Herbal Laboratories [PVT] Ltd. Kochi, Kerala State, India) as an adjuvant in the treatment of patients with type 2 diabetes was carried out during a 3-month period.

**Settings/location:** This study was conducted in two major peripheral clinics of Kuala Lumpur in the Klang Valley, Malaysia.

**Subjects:** A total of 39 Cogent db-treated cases and 40 age-matched controls were recruited for this preliminary study. Nineteen (19) subjects (10 and 9 from control and treatment groups, respectively) dropped out of the study leaving a total of 60 subjects (30 each for control and treatment groups) who completed the study.

**Interventions:** All subjects in the treatment group were given Cogent db (2 tablets three times daily after each meal) in addition to the regular allopathic drugs (Daonil, [Aventis Farma, SA Petaling Jaya, Selangor State, Malaysia] and Diamicron® [Sevier, Bangkok, Thailand], with or without Metformin [Upha Corporation, Bangi, Selangor State, India]) that they took in common with the control group.

**Outcome measures:** Thirty-two (32) clinical variables were investigated, including liver enzymes, kidney function tests, hematologic parameters, blood glucose, insulin, and C-peptide assays.

**Results:** At the end of 3 months it was found that there was a significant decrease in the levels of fasting and postprandial blood glucose, cholesterol, triglycerides, glycated hemoglobin (Hb A<sub>1C</sub>) and fasting insulin in the treatment group compared to the controls. Cogent db did not alter the liver function tests, hematologic parameters, or the kidney function tests.

**Conclusions:** These results concur with earlier animal studies that indicate that Cogent db is safe, reliable, tolerable, and efficacious in the control of type 2 diabetes mellitus.

## INTRODUCTION

Diabetes mellitus is one of the most common chronic diseases, affecting more than 120 million individuals worldwide (McCarty and

Zimmet, 1994). The number is expected to double by the year 2010 because of the rising prevalence, predominantly of non-insulin-dependent diabetes mellitus (type 2 diabetes mellitus) especially in the Asia-Pacific region. In Malaysia,

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diabetes mellitus affects more than 1 million patients, causing severe debilitating conditions (Mustaffa, 1998). Because of the associated risk of stroke, coronary artery diseases, and renal failure, diabetes is ranked among the top five killers in most countries. The complications of diabetes include blindness, ulcers leading to amputation, and terminal kidney disease. Without appropriate prevention and control programs, diabetes will certainly become a formidable health problem in this millennium. Worldwide, considerable advances have been made in the management of diabetes and its complications in recent years. Despite improved understanding of its pathogenesis and the identification of the various risk factors, prevention of the disease and/or its complications still pose formidable health problems.

Traditionally, type 2 diabetes mellitus has been controlled with dietary restriction, exercise, oral hypoglycemic drugs, such as insulin-sensitizers,  $\alpha$ -glucosidase inhibitors, glucagon-like peptide 1, sulphonylureas, and biguanides. These have remained the mainstay of oral hypoglycemic treatment for more than 30 years. However, maintaining glucose control in these patients remains a difficult task (Mustaffa, 1998).

As far back as the fifth and sixth centuries, Indian physicians detected diabetes mellitus from the sweet taste of urine and its attraction of ants, and Ayurvedic medicine is reported to have been successfully used (John, 1997). As an alternative mode of treatment, Ayurvedic medicine has been claimed to be less toxic and more efficacious. One such drug, which has undergone some clinical evaluation in India, is Cogent db (Cybele Herbal Laboratories [PVT] Ltd., Kochi, Kerala State, India), each tablet of which contains the following: *Azadirachta indica* (3.0 g), *Phyllanthus emblica* (0.7 g), *Circuma longa* (0.8 g), *Trigonella foenum graceum* linn (0.1 g), *Syzygium cumini* (2.0 g), *Tribulus terrestris* (1.0 g), *Terminallia bellirica* (0.7 g), *Terminalla chebula* (0.7 g), and *Routula aquatica* (1.0 g) (Joy and Kuttan, 1998).

This herbal preparation has been reported to:

1. Reduce blood glucose to normal levels;
2. Repair damaged pancreatic cells;
3. Reduce or prevent the development of dia-

betic complications, such as retinopathy and nephropathy;

4. Control the development of microangiopathic lesions; and
5. Control other long-term complications of diabetes mellitus (Joy and Kuttan, 1998).<sup>\*†‡</sup>

This study evaluated the efficacy of Cogent db as an adjuvant in the treatment of patients with type 2 diabetes in Malaysia, during a 3-month trial period. This was to gather preliminary data for assessment of the overwhelming anecdotal claims of clinical efficacy of this drug, in a period prior to the development of placebo. The main outcome measures were clinical improvement, blood glucose control, glycosylated hemoglobin (Hb A<sub>1C</sub>), serum insulin, and C-peptide levels. Safety and tolerability of Cogent db was measured by recording all adverse events, laboratory investigations, including hematologic tests, liver enzymes, urea, and creatinine.

## SUBJECTS AND METHODS

### Subjects

Table 1A shows the demographic and clinical characteristics of the 79 subjects initially recruited for this study. The patients were recruited from the peripheral clinics in the Klang Valley. They were from among patients seeking treatment for diabetes mellitus and were recruited after obtaining oral and written consent. Recruitment was free of coercion or inducement. Each subject was assigned to the treatment or control group by a standard randomization process. All patients (both treatment and control groups) have had diabetes for periods ranging

\*Giridharan NV. Effect of Cogent db on mutant obese rats with impaired glucose tolerance (WNIN/GR-ob). Technical report on the project submitted to Cybele Herbal Laboratories [PVT] Ltd., 2000.

†Kamath S, Sharat Kumar N, Singh RG, Usha K. Effect of Cogent db (an herbal formulation) in the treatment of diabetes mellitus. Report submitted to Cybele Herbal Laboratories [PVT] Ltd. 2000.

‡Kuttan R. The anti-diabetic activity of an herbal preparation Cogent db on 10 patients. Clinical trial report submitted to Cybele Herbal Laboratories [PVT] Ltd. 1998.

TABLE 1A. DEMOGRAPHIC AND CLINICAL CHARACTERISTICS OF 79 SUBJECTS AT DAY 0

| Variables                         | Control group n (%) | Treatment group n (%) | Total n (%) |
|-----------------------------------|---------------------|-----------------------|-------------|
| Age                               |                     |                       |             |
| 30–39.9 years                     | 6 (15.0)            | 1 (2.5)               | 7 (8.8)     |
| 40–49.9 years                     | 15 (37.5)           | 15 (38.5)             | 30 (37.9)   |
| 50–59.9 years                     | 13 (32.5)           | 10 (25.6)             | 23 (29.1)   |
| >60 years                         | 6 (15.0)            | 13 (33.3)             | 19 (24.1)   |
| Gender                            |                     |                       |             |
| Male                              | 29 (72.5)           | 23 (58.9)             | 52 (65.8)   |
| Female                            | 11 (27.5)           | 16 (41.0)             | 27 (34.2)   |
| Ethnicity                         |                     |                       |             |
| Indians                           | 22 (55.0)           | 27 (69.2)             | 49 (62.1)   |
| Non-Indians                       | 18 (45.0)           | 12 (30.8)             | 30 (37.9)   |
| Mean weight (kg)                  | 58.2 ± 8.3          | 64.1 ± 6.1            | 62.3 ± 6.2  |
| Day 0                             |                     |                       |             |
| Mean height (in meters)           | 1.54 ± 0.3          | 1.73 ± 0.1            | 1.64 ± 0.1  |
| Mean BMI kg/m <sup>2</sup>        | 26.0 ± 2.5          | 24.1 ± 2.0            | 24.8 ± 1.6  |
| Mean duration of diabetes (years) | 9.1 ± 2.6           | 8.9 ± 2.3             | 8.7 ± 3.6   |

± Standard deviation.  
BMI, body mass index.

from 3 to 15 years. They were either on Daonil (Aventis Farma, SA Petaling Jaya, Selangor State, India) or Diamicon® (Servier, Bangkok, Thailand) with or without Metformin (Upha Corporation, Bangi, Selangor State, India).

Inclusion criteria

Only patients fulfilling all of the following criteria were included: Ages between 30 and 70 years; male or female (i.e., no gender bias); patients diagnosed as type 2 diabetes mellitus with fasting blood glucose (FBG) > 126 mg and random blood glucose (RBG) > 200 mg; patients who have given written informed consent; and patients able to understand and comply with prescribed dietary recommendations and treatment.

Exclusion criteria

Patients were excluded for the following reasons:  
All patients with type 2 diabetes mellitus taking insulin; patients with diastolic blood pressure greater than 100 mg Hg; obesity (body mass index [BMI] > 30) of an endocrine origin other than diabetes mellitus (e.g., Cushing's syndrome); type 1 diabetes; presence of debil-

itating illness; significant neurologic or psychiatric illness; patients taking medications that may affect their mental status; patients with glaucoma; pregnant women; and patients who have taken any other investigational drug within the preceding month.

CLINICAL ASSESSMENT

A brief history covering subject's biodata (name, age, gender, height, weight, BMI, and blood pressure) and current medical history was obtained from every subject who was also required to indicate the presence or otherwise of diabetic symptoms, such as polyuria, polydypsia, numbness, fatigue, and lethargy. After obtaining the baseline data all subjects were required to adhere to a list of prescribed food items (mainly food rich in protein and fibers but low in carbohydrates) and daily physical exercise.  
In addition, 2 hours after a typical Malaysian breakfast (Chinese noodles with tea or coffee) blood was taken for the postprandial glucose and stimulated insulin. This was done in conjunction with the following laboratory investigations at day 0 (i.e., pretreatment) and at the end of 3 months.

Testing

*Urine analysis.* This includes urine biochemistry and urine microscopy. The variables include specific gravity, pH, protein, glucose, ketone, urobilinogen, bilirubin, nitrite, hemoglobin, and leukocyte esterase

*Hematology.* This includes hemoglobin, hematocrit (HCT), red blood cell count (RBC), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), white blood cell count (WBC), platelets and differential counts.

*Clinical chemistry:* This includes glucose, urea, creatinine, albumin, alkaline phosphatase, aspartate transaminase (AST), alanine aminotransferase, total bilirubin Hb A<sub>1C</sub> and total glycated Hb.

*Lipid profile:* This includes triglycerides, total cholesterol, high-density lipoprotein (HDL) cholesterol and low-density lipoprotein (LDL) cholesterol.

*Endocrinology:* This includes fasting and stimulated insulin assay and C-peptide assay.

Duration of treatment

The treatment group took Cogent db (2 tablets after breakfast, lunch, and dinner) for 3 months after the baseline assessment. This was in addition to the ongoing allopathic drugs taken by both control and treatment groups. The allopathic drug regimen ranged between a divided daily intake of Daonil (50–200 mg), Diamicron (80–160 mg), and metformin (1 g–1.5 g) and was maintained throughout the 3 months of the study. Cogent db was supplied by Oze Marketing Pte Ltd., Kuala Lumpur (a registered Malaysian company).

Adverse events

The protocol required that the incidence and severity of all adverse effects or unusual events be recorded in the appropriate forms and their nature, onset, and duration reported to the drug manufacturers.

Statistics

Data analysis was done with the aid of the SPSS statistical software version 9.01. When appropriate, comparisons were made on the basis of the Student's *t*-test with significance accepted at values of *p* < 0.05.

TABLE 1B. DEMOGRAPHIC AND CLINICAL CHARACTERISTICS OF 60 SUBJECTS WHO COMPLETED THE STUDY

| Variables                         | Control group n (%) | Treatment group n (%) | Total n (%) |
|-----------------------------------|---------------------|-----------------------|-------------|
| Age                               |                     |                       |             |
| 30–39.9 years                     | 5 (16.7)            | 1 (3.3)               | 6 (10.0)    |
| 40–49.9 years                     | 13 (43.4)           | 14 (46.7)             | 27 (45.0)   |
| 50–59.9 years                     | 8 (26.7)            | 8 (26.7)              | 16 (26.7)   |
| >60 years                         | 4 (13.3)            | 7 (23.3)              | 11 (18.3)   |
| Gender                            |                     |                       |             |
| Male                              | 21 (70.0)           | 17 (56.7%)            | 38 (63.3)   |
| Female                            | 9 (30.0)            | 13 (43.3%)            | 22 (36.7)   |
| Ethnicity                         |                     |                       |             |
| Indians                           | 20 (66.7)           | 26 (86.7)             | 46 (76.7)   |
| Non-Indians                       | 10 (33.3)           | 4 (13.3)              | 14 (23.3)   |
| Mean weight (kg)                  |                     |                       |             |
| Day 0                             | 61.9 ± 6.4          | 63.8 ± 6.2            | 62.7 ± 7.7  |
| 3 months                          | 62.3 ± 7.2          | 65.1 ± 5.8            | 62.9 ± 5.8  |
| Mean height (in meters)           | 1.58 ± 0.1          | 1.7 ± 0.1             | 1.6 ± 0.08  |
| Mean BMI kg/m <sup>2</sup>        |                     |                       |             |
| Day 0                             | 24.8 ± 2.3          | 22.6 ± 2.9            | 23.8 ± 2.9  |
| 3 months                          | 24.2 ± 1.9          | 23.1 ± 1.7            | 23.9 ± 2.1  |
| Mean duration of diabetes (years) | 8.8 ± 3.3           | 8.4 ± 3.1             | 8.5 ± 4.0   |

± Standard deviation.  
BMI, body mass index.

TABLE 2. SUMMARY OF AGE DISTRIBUTION OF SUBJECTS

| Age                | Control group | Treatment group |
|--------------------|---------------|-----------------|
| Mean               | 47.6          | 50.3            |
| Median             | 43.5          | 48.0            |
| Minimum            | 33.0          | 33.0            |
| Maximum            | 68.0          | 77.0            |
| Range              | 35            | 44              |
| Standard deviation | 9.8486        | 9.9143          |

*p* > 0.05 (no significant difference for any of the values).

RESULTS

Demography

Of the 79 subjects initially recruited, only 60 completed the study. The rest (19) were excluded for reasons of noncompliance with prescribed medication (4 subjects) and/or outright request to withdraw from the study (15 subjects). The withdrawals occurred within the first 5 weeks of the study and none was for reasons of adverse drug reactions. Those who dropped out were predominantly non-Indians and they gave reasons suggestive of a lack of confidence in the Ayurvedic prescription. Table 1B shows the demographic and clinical characteristics of the 60 subjects who completed the study. Of the 60, the majority (45%) were within the 40–49.9 year age group. This was closely followed by the 50–59.9 age group (27%); both groups comprised 72% of all the subjects. There were 38 males and 22 females. Indians comprised 76.7% of the total, compared to 23.3% of non-Indians (Malays 21.7% and Chinese 1.6%). The unequal race distribution may be because Indians believe more in Ayurvedic medicine than the other races and

also possibly because most Indians sought treatment in the peripheral private clinics. This may also explain the predominance of non-Indians among the 19 subjects who withdrew from the study.

The data in Table 2 show the statistical summary of age distribution for control and treatment groups. The mean ages were 47.6 and 50.3 years, respectively, for the control and treatment groups. The minimum age in either control or treatment group was 33 years while the maximum was 68 and 77 years, respectively. The data show that there was no significant difference in the age distribution between the control and treatment groups.

Presenting complaints and diabetes severity grading

Table 3 shows the frequency distribution of the presenting complaints of the subjects on recruitment and after 3 months of treatment. The common complaints were polyuria, polyphagia, polydipsia, fatigue/lethargy, and neuromuscular complaints including numbness, pain in the extremities, discoloration of the skin, slow healing of wounds, loss of sensation in the feet, bladder and bowel discomfort, and visual disturbances. The pattern of complaints after 3 months shows that there was a marked improvement in patients treated with Cogent db. Treatment with Cogent db relieved polyuria, fatigue, and lethargy in all patients who complained of these symptoms at day 0. There was a marked reduction in the numbers of the patients with other systemic complaints. Of the 7 (23.3%) who complained of polyphagia, only 2 (6.7%) still manifested this symptom

TABLE 3. SUMMARY OF SUBJECTS' COMPLAINTS

| Clinical findings        | Before treatment n (%) |                 | After treatment n (%) |                     |
|--------------------------|------------------------|-----------------|-----------------------|---------------------|
|                          | Control group          | Treatment group | Control group         | Treatment group     |
| Polyuria                 | 18 (60)                | 16 (53)         | 18 (60)               | 0 (0) <sup>a</sup>  |
| Polyphagia               | 9 (30)                 | 7 (23)          | 9 (30)                | 2 (7) <sup>a</sup>  |
| Polydipsia               | 16 (53)                | 18 (60)         | 16 (53)               | 1 (3) <sup>a</sup>  |
| Fatigue and lethargy     | 27 (90)                | 30 (100)        | 27 (90)               | 0 (0) <sup>a</sup>  |
| Neuromuscular complaints | 23 (77)                | 19 (63)         | 23 (77)               | 5 (17) <sup>a</sup> |

<sup>a</sup>Frequency distribution of symptoms as reported by the subjects before (day 0) and after (3 months) treatment. No changes occurred in the control group, but significant drops in frequency was recorded for all the complaints in the Cogent db-treated group.

TABLE 4A. DISTRIBUTION OF PATIENTS ACCORDING TO DIABETES SEVERITY GRADES  
BASED ON FBG BEFORE (AT DAY 0) AND AFTER (3 MONTHS) TREATMENT

| Grade | FBG<br>(mg%) | Before treatment     |                           |              | After treatment      |                           |              |
|-------|--------------|----------------------|---------------------------|--------------|----------------------|---------------------------|--------------|
|       |              | Control<br>group (%) | Treatment<br>group<br>(%) | Total<br>(%) | Control<br>group (%) | Treatment<br>group<br>(%) | Total<br>(%) |
| I     | <100         | 0 (0)                | 0 (0)                     | 0 (0)        | 0 (0)                | 0 (0)                     | 0 (0)        |
| II    | 100–150      | 6 (20.0)             | 11 (30.7)                 | 17 (28.3)    | 5 (16.7)             | 13 (43.3)                 | 18 (30.0)    |
| III   | 150–200      | 8 (26.7)             | 7 (23.3)                  | 15 (25.0)    | 7 (23.3)             | 12 (40.0)                 | 19 (31.7)    |
| IV    | ≥200         | 16 (53.3)            | 12 (40.0)                 | 28 (46.7)    | 18 (60.0)            | 5 (16.7)                  | 23 (38.3)    |
| Total |              | 30 (100)             | 30 (100)                  | 60 (100)     | 30 (100)             | 30 (100)                  | 60 (100)     |

FBG, fasting blood glucose.

TABLE 4B. DISTRIBUTION OF PATIENTS ACCORDING TO DIABETES SEVERITY GRADES  
BASED ON PBG BEFORE (AT DAY 0) AND AFTER (3 MONTHS) TREATMENT

| Grade | PBG<br>(mg%) | Before treatment     |                           |              | After treatment      |                           |              |
|-------|--------------|----------------------|---------------------------|--------------|----------------------|---------------------------|--------------|
|       |              | Control<br>group (%) | Treatment<br>group<br>(%) | Total<br>(%) | Control<br>group (%) | Treatment<br>group<br>(%) | Total<br>(%) |
| I     | <150         | 0 (0)                | 0 (0)                     | 0 (0)        | 0 (0)                | 3 (10.0)                  | 3 (5.0)      |
| II    | 150–225      | 3 (10.0)             | 10 (10.0)                 | 13 (21.6)    | 3 (10.0)             | 15 (50.0)                 | 18 (30.0)    |
| III   | 225–300      | 13 (43.3)            | 12 (40.0)                 | 25 (41.7)    | 9 (30.0)             | 9 (30.0)                  | 18 (30.0)    |
| IV    | ≥300         | 14 (46.7)            | 8 (26.7)                  | 22 (36.7)    | 18 (60.0)            | 3 (10.0)                  | 21 (35.0)    |
| Total |              | 30 (100)             | 30 (100)                  | 60 (100)     | 30 (100)             | 30 (100)                  | 60 (100)     |

PBG, postprandial blood glucose.

at 3 months while only 1 (3.3%) of the 18 (60.0%) who complained of polydipsia still had this symptom. Of the 19 (63.3%) cases who had neuromuscular complaints, only 5 (16.7%) continued to have these complaints after 3 months of treatment with Cogent db. None of the patients complained of any side-effects except for a mild diarrhea that lasted for a week in 4 patients. No other adverse effect was reported. In the control group, however, there were no changes in symptomatology after 3 months.

The severity of disease was determined by FBG or postprandial blood glucose (PBG) levels and graded as shown in Tables 4A and 4B. Table 4A shows the gradation based on FBG levels. At the onset of the study (upper panel), 17 patients (28.3 %) were in the grade II category with 11 of these in the treatment group and 6 in the control. In the grade III category, a total of 15 subjects (25.0%), 7 and 8 of whom were in the treatment and control groups, respectively. A total of 28 patients (46.7%) were in grade IV, 12 of these were in the treatment

group and 16 in the control. No subjects were in the grade I category.

None of the 60 subjects who completed the study were in the grade I category after 3 months of treatment (Table 4A), whereas 18 (30.0%), 19 (31.7%), and 23 (38.3%) patients were within grades II, III, and IV categories, respectively. Most patients in the Cogent db treatment group were either in the grade II or grade III category, with 13 (43.3%) and 12 (40.0%) patients, respectively. The number of cases in grade IV under the treatment group decreased from 12 (40.0%) at day 0 (upper panel) to 5 (16.7%) at 3 months (lower panel). This represents a 58.3% decrease in the number of patients with this level of severity of diabetes. However, the grade III and II categories recorded 71.4% and 18.2% increases, representing a change from 7 to 12 and 11 to 13, respectively. These increases are attributable to the decreases that were observed in the grade IV category. The general tendency to decrease to lower grades of severity after treatment with

Cogent db was not observed in the control group.

Table 4B shows the diabetes severity classification based on the PBG levels. At day 0 (upper panel) a total of 13 patients (21.6%) were in the grade II category with 10 of these in the treatment group and 3 in the control. In the grade III category, a total of 25 (41.7%) subjects were recorded, 12 and 13 of whom were in the treatment and control groups, respectively. A total of 22 patients (36.7%) were in grade IV, 8 of these in the treatment group and 14 in the control. No subjects were in the grade I category. After 3 months of treatment (lower panel) however, only 3 of the 8 patients in the treatment group who started off at grade IV severity remained in that category, a 62.5% decrease. The rest have become healthier. Of the 12 in grade III severity there was a 25.0% decrease to only 9, whereas in the grade II category the number of patients increased by 50.0%, from 10 to 15. At day 0 (upper panel) none of the subjects were within grade I severity, but at 3 months, 3 (10.0%) of the Cogent db-treated patients improved into this grade. At day 0, there were 14 patients in the control group who were within grade IV severity. This number increased by 28.6% to 18 at 3 months. Patients of grade III severity decreased by 30.7%, from 13 to 9 subjects while the number in grade II remained unchanged at 3. No patient in the control group was in the grade I category either at day 0 or at 3 months. The general trend was that patients in the treatment group moved from higher to lower grades of severity. It is interesting to note that 10% of patients in this group moved into grade I. In the control group, however, the reverse was true as typified by

the 28.6% increase in the number of patients in grade IV.

Urine glucose level

Table 5 shows the urine glucose levels of the control and the treated subjects at day 0 and after 3 months of treatment. At day 0, 19 (31.7%) of the total of 60 subjects had urine glucose levels within the normal limits. The rest (41; 68%) had higher than normal urine glucose levels ranging from 1+ to 4+, with 19 (31.7%) subjects in the 4+ ( $\geq 56$  mmol/L) range and 9 (15.0%), 10 (16.7%), and 3 (5.0%) in the 1+, 2+, and 3+ ranges, respectively. After 3 months of treatment the number of patients with normal urine glucose increased by 37% from 19 to 26. This means that those with abnormal urine glucose levels declined from 41 to 34. In the control group, there were only minor changes: 6 had normal urine glucose at day 1, and at 3 months this increased to only 7, implying that the figures for those with abnormal urine glucose dropped by only 1 from 24 at day 0 to 23 at 3 months. In the treatment group, subjects with normal urine glucose levels increased from 13 (at day 0) to 19 at 3 months. This reflects a decrease from 17 to 11, in the number of those with abnormal urine glucose in this group.

Urine protein level

Table 6 shows the urine protein biochemistry in the control and treatment groups at day 0 and at 3 months after treatment. Of the total of 60 subjects, 18 (30%) and 42 (70%) recorded normal and abnormal values, respectively, at day 0. Of the abnormal values, 33, 4, and 5 sub-

TABLE 5. DISTRIBUTION OF URINE GLUCOSE LEVELS BEFORE (AT DAY 0) AND AFTER (3 MONTHS) TREATMENT

| Urine glucose level    | Before treatment  |                     |           | After treatment   |                     |           |
|------------------------|-------------------|---------------------|-----------|-------------------|---------------------|-----------|
|                        | Control group (%) | Treatment group (%) | Total (%) | Control group (%) | Treatment group (%) | Total (%) |
| Normal                 | 6 (20)            | 13 (43)             | 19 (32)   | 7 (23)            | 19 (63)             | 26 (43)   |
| 1+ ( $\geq 3$ mmol/L)  | 6 (20)            | 3 (10)              | 9 (15)    | 5 (17)            | 2 (7)               | 7 (12)    |
| 2+ ( $\geq 6$ mmol/L)  | 6 (20)            | 4 (13)              | 10 (17)   | 6 (20)            | 4 (13)              | 10 (17)   |
| 3+ ( $\geq 18$ mmol/L) | 2 (7)             | 1 (3)               | 3 (5)     | 0 (0)             | 3 (10)              | 3 (5)     |
| 4+ ( $\geq 56$ mmol/L) | 10 (33)           | 9 (30)              | 19 (32)   | 12 (40)           | 2 (7)               | 14 (23)   |
| Total                  | 30 (100)          | 30 (100)            | 60 (100)  | 30 (100)          | 30 (100)            | 60 (100)  |

TABLE 6. DISTRIBUTION OF URINE PROTEIN LEVELS BEFORE (DAY 0) AND AFTER (3 MONTHS) TREATMENT<sup>a</sup>

| Urine protein grades | Before treatment  |                     |           | After treatment   |                     |           |
|----------------------|-------------------|---------------------|-----------|-------------------|---------------------|-----------|
|                      | Control group (%) | Treatment group (%) | Total (%) | Control group (%) | Treatment group (%) | Total (%) |
| Normal               | 10 (33)           | 8 (27)              | 18 (30)   | 7 (23.3)          | 17 (56.7)           | 24 (40.0) |
| 1+ (0.25 g/L)        | 16 (53)           | 17 (57)             | 33 (55)   | 16 (53.3)         | 9 (30.0)            | 25 (41.7) |
| 2+ (0.75 g/L)        | 2 (7)             | 2 (7)               | 4 (7)     | 3 (10.0)          | 2 (6.7)             | 5 (8.3)   |
| 3+                   | 2 (7)             | 3 (10)              | 5 (8)     | 4 (13.3)          | 2 (6.7)             | 6 (10.0)  |
| Total                | 30 (100)          | 30 (100)            | 60 (100)  | 30 (100)          | 30 (100)            | 60 (100)  |

<sup>a</sup>Urine protein was measured by the modified pyrogallol red molybdate adaptation method.

jects, respectively, recorded 1+, 2+, and 3+ levels of proteinuria.

After 3 months of treatment there was a significantly ( $p < 0.001$ ) higher change toward normal in the treatment group compared to the control. A total of 24 and 36 subjects, respectively, recorded normal and abnormal values. Of the normal values, 7 were in the control group (reflecting a decline of 3 from 10 at day 0) and 17 in the treatment group (reflecting an increase of 9 (>100%) from day 0). Of the abnormal cases, 16 control subjects recorded 1+ as was the case at day 0, while 7 subjects had values at 2+ and above, a 75% increase from the 4 at day 0. In the treatment group, 9 subjects recorded 1+ urine protein, an 88% decrease from the 17 subjects in this category at day 0, while 4 subjects had values at 2+ and above compared to 5 subjects at day 0, a 20% decline.

Laboratory tests

Table 7A shows the baseline group mean values for each of the laboratory tests done at day 0. Thirty-two (32) variables were examined. All parameters, including hematologic and liver function tests, were found to be within the normal levels in both the control and treatment groups, except for the triglycerides, total cholesterol, FBG, PBG, Hb A<sub>1C</sub>, total glycated hemoglobin, stimulated insulin, fasting C-peptide, and stimulated C-peptide values, which were all higher than normal in either control or treatment group.

Triglycerides and total cholesterol were elevated in both control and treatment groups. The mean FBG levels of the control and treat-

ment groups were 10.5 and 12.7 mmol/L, respectively. This shows that in spite of allopathic medication for at least 3 months, blood glucose levels were not well controlled in all the patients. The Hb A<sub>1C</sub> levels of 9.5% and 10.2% for the control or treatment group, respectively was elevated much above the normal limits (4.40%–6.40%). Similarly, the glycated hemoglobin values of 10.9 and 11.8 for the control and treatment groups, respectively, were much higher than the normal limits (4.8%–7.8%), indicating a poor glycemic control in the preceding 3–4 months.

All subjects registered normal values (0.0–17.5 μU/mL) for fasting insulin assay, except for 12 (20%) who had values ranging from 18.6 μU/mL–23.9 μU/mL. Stimulated insulin mean values of 40.7 and 52.0 μU/mL, respectively, for the control and treatment groups, indicate that insulin was being secreted.

Table 7B shows the mean values of the various laboratory results from the treatment and control groups after 3 months. In both groups the serum albumin, bilirubin, and bile acids were within normal limits, suggesting that neither Cogent db nor the control treatment regimen caused any significant hepatocellular damage within the period of this study. In the control group, however, the AST levels increased by 50.7% above the baseline value of 26.3 IU/L (Table 7A) to 53.4 IU/L, thus exceeding the upper limit of normal (15–37 IU/L). This suggests some degree of hepatocellular damage. With the treatment group, the AST baseline levels (29.30 IU/L) increased by 15.8% to 33.95 IU/L, still within the normal range and significantly ( $p < 0.002$ ) lesser than the control group.



TABLE 7A. MEAN VALUES OF LABORATORY TESTS BEFORE TREATMENT (DAY 0)

| Variable                                         | Control ± SD | Treatment ± SD | Normal range                    |
|--------------------------------------------------|--------------|----------------|---------------------------------|
| Alanine aminotransferase                         | 49.8 ± 28.8  | 57.4 ± 29.9    | 30–65 IU/L                      |
| Albumin                                          | 38.3 ± 6.4   | 39.9 ± 3.8     | 35–50 g/L                       |
| Alkaline Phosphatase                             | 99.3 ± 31.8  | 99.5 ± 33.1    | 50–136 IU/L                     |
| Aspartate aminotransferase (AST)                 | 26.3 ± 19.8  | 29.3 ± 19.1    | 15–37 IU/L                      |
| Creatinine                                       | 82.9 ± 36.6  | 88.9 ± 29.9    | 70–115 μmol/L                   |
| Total bilirubin                                  | 12.6 ± 18.9  | 13.3 ± 7.9     | 3.4–17 μmol/L                   |
| Urea                                             | 4.9 ± 2.7    | 4.7 ± 1.3      | 2.5–6.4 mmol/L                  |
| Triglycerides                                    | 2.3 ± 1.2    | 2.3 ± 1.1      | 0.45–1.82 mmol/L                |
| Total cholesterol                                | 5.7 ± 1.0    | 5.7 ± 1.3      | 3.6–5.2 mmol/L                  |
| Low-density lipoproteins (LDL)                   | 3.7 ± 1.1    | 3.6 ± 1.4      | 1.68–4.53 mmol/L                |
| High-density lipoproteins (HDL)                  | 1.1 ± 0.3    | 1.2 ± 1.1      | 1.10–2.20 mmol/L                |
| Mean corpuscular hemoglobin (MCH)                | 27.8 ± 2.3   | 28.3 ± 1.0     | 27.0–32.0 pg                    |
| Mean corpuscular hemoglobin concentration (MCHC) | 335.0 ± 7.3  | 330.4 ± 19.7   | 300–350 g/L                     |
| Mean corpuscular volume (MCV)                    | 82.6 ± 5.9   | 84.7 ± 3.1     | 76–96 fl                        |
| Hematocrit                                       | 0.4 ± 5.2    | 0.4 ± 4.9      | 0.37–0.47                       |
| Hemoglobin                                       | 142.8 ± 17.1 | 147.0 ± 15.4   | 115.0–165.0 g/L                 |
| % Eosinophils                                    | 4.0 ± 1.9    | 2.8 ± 1.3      | 1–6%                            |
| % Lymphocytes                                    | 34.5 ± 7.1   | 37.2 ± 7.8     | 20–45%                          |
| % Monocytes                                      | 6.6 ± 2.1    | 7.2 ± 2.3      | 2–10%                           |
| % Basophils                                      | 1.0 ± 0.0    | 1.0 ± 0.0      | 0–1%                            |
| % Neutrophils                                    | 54.5 ± 7.4   | 50.9 ± 9.6     | 40–75%                          |
| Platelets                                        | 246.8 ± 68.8 | 261.4 ± 67.6   | 150–400 (10 <sup>9</sup> /L)    |
| Red blood cells (RBC)                            | 5.2 ± 0.5    | 5.3 ± 0.6      | 3.80–5.80 (10 <sup>12</sup> /L) |
| White blood cells (WBC)                          | 8.0 ± 1.6    | 9.2 ± 2.3      | 4.0–11.0 (10 <sup>9</sup> /L)   |
| Fasting blood glucose (FBG)                      | 10.5 ± 3.9   | 12.7 ± 4.9     | 3.9–6.1 mmol/L                  |
| Postprandial glucose (PBG)                       | 14.9 ± 4.7   | 18.2 ± 6.6     | 3.9–6.1 mmol/L                  |
| HBA <sub>1c</sub>                                | 9.5 ± 2.2    | 10.2 ± 3.1     | 4.40%–6.40%                     |
| Total glycated hemoglobin                        | 10.9 ± 2.7   | 11.8 ± 3.7     | 4.80%–7.80%                     |
| Fasting insulin                                  | 13.8 ± 9.9   | 11.5 ± 9.4     | 0.0–17.5 μU/mL                  |
| Stimulated Insulin                               | 40.7 ± 31.7  | 52.0 ± 71.7    | 0.0–17.5 μU/mL                  |
| Fasting C-peptide                                | 3.2 ± 1.6    | 5.5 ± 12.8     | 0.6–2.8 ng/mL                   |
| Stimulated C-peptide                             | 7.7 ± 4.3    | 7.1 ± 5.3      | 0.6–2.8 ng/mL                   |

± Standard deviation (SD).

In general only marginal changes were recorded between day 0 and 3 months in the triglycerides and total cholesterol levels (see Tables 7A and 7B). In the control group, triglycerides level rose by 8.7%, from 2.29 to 2.49 mmol/L, whereas total cholesterol increased by 4.9%, from 5.7 to 6.0 mmol/L. In the treatment group, triglycerides levels decreased by 12.9%, from 2.3 to 2.0, significantly ( $p < 0.006$ ) lower than control whereas cholesterol decreased by 7.7%, from 5.7 to 5.3 mmol/L, significantly ( $p < 0.001$ ) less than control.

In the treatment group FBG decreased significantly ( $p < 0.043$ ) by 27.9% from 12.2 to 8.8 mmol/L, compared to the control group, which recorded a 17.3% increase from 10.5 to 12.3

mmol/L. The levels exceeded the upper limit of normal FBG (3.9–6.1 mmol/L) although the treatment group showed a tendency toward decreasing to normal. The PBG levels dropped significantly ( $p < 0.009$ ) by 34% (from 18.2 mmol/L at day 0 to 11.9 mmol/L at 3 months) in the treatment group compared to the increase by 18.6% from 14.9 mmol/L to 18.3 mmol/L in the control group. The results show that Cogent db can reduce blood glucose levels within a 3-month period.

The mean Hb A<sub>1c</sub> levels in the treatment group dropped from the day 0 level of 10.2% to 8.3% at 3 months, significantly ( $p < 0.001$ ) different from the control where an increase from 9.5% to 10.4% was recorded (normal

TABLE 7B. MEAN VALUES OF LABORATORY TESTS AFTER TREATMENT (3 MONTHS)

| Variable                                         | Control group $\pm$ SD | Treatment group $\pm$ SD | Normal range              |
|--------------------------------------------------|------------------------|--------------------------|---------------------------|
| Alanine aminotransferase                         | 30.8 $\pm$ 22.3        | 35.1 $\pm$ 16.1          | 30–65 IU/L                |
| Albumin                                          | 41.2 $\pm$ 4.5         | 40.3 $\pm$ 2.9           | 35–50 g/L                 |
| Alkaline Phosphatase                             | 99.1 $\pm$ 34.5        | 96.5 $\pm$ 31.2          | 50–136 IU/L               |
| Aspartate aminotransferase (AST)                 | 53.4 $\pm$ 20.9        | 33.9 $\pm$ 17.2          | 15–37 IU/L                |
| Creatinine                                       | 96.5 $\pm$ 31.5        | 81.5 $\pm$ 16.2          | 70–115 $\mu$ mol/L        |
| Total bilirubin                                  | 14.5 $\pm$ 8.8         | 11.2 $\pm$ 4.9           | 3.4–17 $\mu$ mol/L        |
| Urea                                             | 4.9 $\pm$ 1.2          | 4.6 $\pm$ 1.6            | 2.5–6.4 mmol/L            |
| Triglycerides                                    | 2.5 $\pm$ 1.1          | 2.0 $\pm$ 0.9            | 0.45–1.82 mmol/L          |
| Total cholesterol                                | 6.0 $\pm$ 1.4          | 5.3 $\pm$ 0.7            | 3.6–5.2 mmol/L            |
| Low-density lipoproteins (LDL)                   | 3.9 $\pm$ 1.6          | 3.6 $\pm$ 0.8            | 1.68–4.53 mmol/L          |
| High-density lipoproteins (HDL)                  | 1.1 $\pm$ 0.3          | 1.1 $\pm$ 0.3            | 1.10–2.20 mmol/L          |
| Mean corpuscular hemoglobin (MCH)                | 28.6 $\pm$ 1.5         | 27.9 $\pm$ 2.5           | 27.0–32.0 pg              |
| Mean corpuscular hemoglobin concentration (MCHC) | 335.9 $\pm$ 7.7        | 336.4 $\pm$ 9.2          | 300–350 g/L               |
| Mean corpuscular volume (MCV)                    | 84.9 $\pm$ 4.3         | 83.3 $\pm$ 6.4           | 76–96 fl                  |
| Hematocrit                                       | 0.5 $\pm$ 8.3          | 0.4 $\pm$ 6.6            | 0.37–0.47                 |
| Hemoglobin                                       | 148.4 $\pm$ 16.5       | 143.5 $\pm$ 20.2         | 115.0–165.0 g/L           |
| % Eosinophils                                    | 2.4 $\pm$ 1.6          | 4.1 $\pm$ 1.9            | 1–6%                      |
| % Lymphocytes                                    | 39.8 $\pm$ 11.3        | 35.7 $\pm$ 7.9           | 20%–45%                   |
| % Monocytes                                      | 6.8 $\pm$ 2.3          | 7.3 $\pm$ 6.5            | 2%–10%                    |
| % Basophils                                      | 1.1 $\pm$ 0.3          | 1.1 $\pm$ 0.5            | 0%–1%                     |
| % Neutrophils                                    | 49.6 $\pm$ 12.6        | 64.0 $\pm$ 53.4          | 40%–75%                   |
| Platelets                                        | 272.1 $\pm$ 84.9       | 248.6 $\pm$ 69.6         | 150–400 ( $10^9$ /L)      |
| Red blood cells                                  | 5.3 $\pm$ 0.6          | 5.2 $\pm$ 0.6            | 3.80–5.80 ( $10^{12}$ /L) |
| White blood cells                                | 13.9 $\pm$ 17.9        | 8.4 $\pm$ 1.7            | 4.0–11.0 ( $10^9$ /L)     |
| Fasting blood glucose (FBG)                      | 12.3 $\pm$ 4.8         | 8.8 $\pm$ 2.2            | 3.9–6.1 mmol/L            |
| Postprandial glucose (PBG)                       | 18.3 $\pm$ 5.5         | 11.9 $\pm$ 3.0           | 3.9–6.1 mmol/L            |
| H B A <sub>1c</sub>                              | 10.4 $\pm$ 2.9         | 8.3 $\pm$ 1.4            | 4.40%–6.40%               |
| Total glycated hemoglobin                        | 12.4 $\pm$ 3.4         | 9.5 $\pm$ 1.6            | 4.80%–7.80%               |
| Fasting insulin                                  | 14.2 $\pm$ 10.8        | 11.3 $\pm$ 7.3           | 0.0–17.5 $\mu$ U/mL       |
| Stimulated insulin                               | 55.8 $\pm$ 73.9        | 38.9 $\pm$ 25.9          | 0.0–17.5 $\mu$ U/mL       |
| Fasting C-peptide                                | 3.7 $\pm$ 2.5          | 3.5 $\pm$ 1.7            | 0.6–2.8 ng/mL             |
| Stimulated C-peptide                             | 7.3 $\pm$ 5.1          | 8.1 $\pm$ 3.9            | 0.6–2.8 ng/mL             |

$\pm$  Standard deviation (SD).

value, 4.4–6.4%). The average total glycated hemoglobin (normal value, 4.8%–7.8%) dropped significantly ( $p < 0.0001$ ) from 11.8% to 9.5% in the treatment group compared to the increase from 10.9% to 12.4% in the control group. At 3 months, the fasting insulin levels remained unaltered in both treatment and control groups. The day 0 average for the treatment group was 11.53  $\mu$ U/mL while it was 11.33  $\mu$ U/mL at 3 months. The figures for the control group were 13.83  $\mu$ U/mL and 14.16  $\mu$ U/mL, respectively, all of which fell within the normal range (0.0–17.5  $\mu$ U/mL). Stimulated insulin levels, however, showed a different pattern. In the control, it increased by 37% from 40.7 to 55.8  $\mu$ U/mL while in the treatment group the values decreased by 25% from 52.0 to 38.9  $\mu$ U/mL.

The average fasting C-peptide levels re-

mained almost unchanged in the control group with 3.2 ng/mL at day 0 and 3.7 ng/mL at 3 months, whereas in the treatment group it decreased significantly ( $p < 0.025$ ) compared to the control, by 37.4%, from 5.5 ng/mL to 3.5 ng/mL. With the treatment group, postprandial C-peptide levels showed a slight (13.5%) increase from 7.1 ng/mL baseline to 8.1 ng/mL at 3 months. In the control group the values are 7.7 ng/mL baseline and 7.3 ng/mL at 3 months.

## DISCUSSION

A number of allopathic drugs have been introduced into the market for the control and prevention of type 2 diabetes mellitus. In spite of these, the prevalence of diabetes increases

each year. Traditional/alternative medicines are gaining popularity in the management of diabetes mellitus, partly because of claims of low side-effects. One such herbal medicine is the Ayurvedic preparation, Cogent db. This medicine has been claimed to rejuvenate the ailing pancreas and to restore its function (Joy and Kuttan, 1998).\*

Animal studies had earlier revealed that oral administration of Cogent db reduced blood glucose levels in normal rats and those made diabetic by alloxan injection or loaded with glucose (Joy and Kuttan, 1998). The lowering of the blood glucose occurred within 12 hours of administration of the drug at a concentration of 100 mg/kg of body weight. Similarly, a correction of impaired glucose tolerance was obtained in WNIN/GR-Ob rats administered 100 mg/kg of Cogent db by Giridharan.\*

The animal studies were followed by various human clinical trials in India, notably by Joy and Kuttan (1998) who found that the drug reduced the FBG after 1 month of treatment of 10 patients with confirmed diabetes. Similar results were also obtained with postprandial glucose. In a much larger trial involving 124 patients (118 of whom had type-2 diabetes), Kamath et al. (2000) showed significant reductions in the FBG and PBG after 4 months of treatment with Cogent db. They confirmed symptomatic and biochemical improvement in all patients. In another study conducted at the Khartoum University, Sudan,<sup>§</sup> there was a significant reduction in the FBG and PBG levels after 3 months in 30 patients treated with Cogent db. In a randomized and double-blind clinical trial carried out at the Cybele Laboratory, India, 100 patients with diabetes were treated with Cogent db for a period of 3 months<sup>||</sup>; 24% of the patients had either grade

III or grade IV diabetes, 43% had grade II, and 4.4% had grade I diabetes. After 3 months of treatment 77% had their blood glucose levels under control while 23% did not. Most of the patients who benefited were those from grade III and grade IV. The results of this study are consistent with these findings in that after 3 months of treatment with Cogent db, there was a significant reduction in FBG ( $p < 0.043$ ) and PBG ( $p < 0.009$ ) levels and shifts of the subjects from higher to lower diabetes severity grades.

At day 0, 12 (40%) of the 30 treated subjects had FBG levels of the grade IV category. This number decreased to 5 (16.7%) at 3 months, thus increasing the number of patients with grade III diabetes from 7 (23%) at day 0 to 12 (40%) at 3 months. Patients with grade II diabetes increased from 11 (31%) at day 0 to 13 (43%) at 3 months. These data indicate that Cogent db-treated subjects tended to improve in their diabetic status, thus shifting from a higher to a lower severity grade. In contrast, in the control group, only minor changes were observed. Of the 30 subjects in this category 16 (53%) were under grade IV severity at day 0 with the figure increasing to 18 (60%) at 3 months. Eight (8) (27%) and 6 (20%) subjects, respectively, were in grades III and II at day 0 and the corresponding figures at 3 months was 7 (23%) and 5 (17%), respectively, showing a tendency to shift toward higher severity grades. No subject in the treatment group was in grade I, either at day 0 or at 3 months. The PBG levels in the treatment group also showed significant shifts toward lower severity grades with patients in grade IV decreasing from 8 (27%) at day 0 to 3 (10%) at 3 months. Subjects in grade III decreased from 12 (40%) to 9 (30%) while those in grade II made up for these decreases by increasing from 10 (33%) to 15 (50%) and from 0 to 3 (10%) subjects in grade I. In the control group, grade IV subjects increased from 14 (47%) to 18 (60%) while the grade III made up for this by a decrease from 13 (43%) to 9 (30%) subjects. There was no change in the number (3) of patients in grade II and none progressed to grade I.

The most interesting change to note is that there was a definite significant shift of subjects from a high to a lower severity grade in the treatment compared to the control group

<sup>§</sup>El-Toum MA. Anti-diabetic activity of Cogent db tablets on 30 patients in Khartoum, Sudan. Clinical Report submitted to Cybele Herbal Laboratories Pte. Ltd. 2000.

<sup>||</sup>Cybele. A randomized clinical trial on 100 diabetic patients treated with Cogent db. Clinical Report submitted to Cybele Herbal laboratories Pte. Ltd. 2000.

<sup>¶</sup>Cybele. The utilization of Cogent db on the treatment of 80 diabetic patients. Clinical report submitted to Cybele Herbal Laboratories Pte. Ltd. 2000.

where the opposite trend was observed. This confirms the ability of Cogent db to reduce blood glucose levels.

The study by Joy and Kuttan (1998) showed that results of liver function tests in the rats were not altered after Cogent db treatment. Administration of the drug reduced the high bilirubin levels, serum glutamate pyruvate, and serum lipid peroxidase. Similar results were obtained in rat<sup>\*</sup> and human studies.<sup>§</sup> In the present study, except for the significant rise in AST level in the control group, all the liver enzymes remained within normal limits after 3 months in both the control and the treatment group. The results indicate that within the 3-month period, Cogent db was not hepatotoxic and, therefore, relatively safe. It is remarkable that AST levels within the control group rose significantly from 26.3 IU/L at day 0 to 53.4 IU/L at 3 months whereas with the Cogent db-treated group it rose insignificantly from 29.3 to 33.9 IU/L. These data suggest some degree of hepatocellular damage in the control, but not in the treatment group, further indicating that Cogent db may have a protective effect on the liver.

Triglyceride and total cholesterol levels were slightly higher than normal values in both control and treatment groups at day zero (Table 7A). After 3 months of treatment, however, the values tended to increase in the control group, whereas they decreased in the Cogent db-treated group with triglyceride value (2.02 mmol/L) only slightly higher than the upper (1.82 mmol/L) limit of normal ( $p < 0.001$ ) while total cholesterol fell from 5.7 mmol/L at day 0 to the upper limit of normal 5.2 mmol/L at 3 months ( $p < 0.001$ ). This trend is consistent with findings in animal models (Joy and Kuttan, 1998), including the study by Giridharan<sup>\*</sup> who concluded that a low-dose regimen of 100 mg/kg of body weight of Cogent db was effective in controlling cholesterol and triglycerides. It is noteworthy that in this study the triglyceride values returned to almost normal levels at 3 months, suggesting that Cogent db might have some triglyceride-lowering property.

Cogent db has been reported to decrease the levels of glycated hemoglobin in both animal (Joy and Kuttan, 1998) and human<sup>‡</sup> studies. In this study, Cogent db significantly reduced the

level of glycated hemoglobin and the Hb A<sub>1C</sub>, indicating that it has the potential to prevent or delay the complications of diabetes such as heart disease, blindness, nerve and kidney damage. The shift of stimulated insulin and PBG toward normal with Cogent db treatment suggests that Cogent db possesses a normalizing effect on these two parameters.

As with animal (Joy & Kuttan, 1998)<sup>\*</sup> and earlier human studies,<sup>‡</sup> this study indicates that Cogent db does not alter hematologic values such as hemoglobin, total count, and erythrocyte sedimentation rate (ESR), all of which remained within the normal ranges, at day 0 and after 3 months of treatment. This suggests that the drug is safe.

Joy and Kuttan (1998) showed that the blood urea nitrogen was significantly reduced in rats treated with Cogent db. The serum creatinine levels were found to be marginally higher in the case of animals treated with alloxan. This was partially reduced by administration of the herbal extract. The Khartoum study<sup>§</sup> showed that Cogent db does not affect the kidneys because the blood urea and creatinine levels before and after treatment were not altered. In this study, all kidney function tests revealed no abnormalities suggesting a relative safety of the drug with regards to nephrotoxicity.

Joy and Kuttan (1998) found that urinary sugar values also decreased in all 10 patients they treated with Cogent db. In this study, 43% of the Cogent db-treated subjects had normal urine glucose levels at day 0. This figure increased to 63% at day 90 compared to the control with 20% at day 0 and 23% at day 90. These suggest that Cogent db has a protective effect. The same feature was also observed in the gradation of proteinuria: 57% of the Cogent db-treated subjects had normal urine protein profile at 3 months compared to 27% at day 0. The figures for the control group were 23% at 3 months and 33% at day 0. These data suggest a high prevalence of diabetic nephropathy among the subjects in this study, not unusual for those with diabetes in this region (Mustaffa, 1999). The reduction of urine protein levels at 3 months, especially in the treatment group, supports earlier claims (Joy and Kuttan, 1998)<sup>\*‡</sup> that Cogent db prevents the diabetic complication of nephropathy. Complications such as ketonuria, hemoglobinuria, and the presence of leukocyte esterase

in the urine were neither detected at the beginning nor at the end of 3 months.

In agreement with a more recent study,<sup>11</sup> polyurea, fatigue, and lethargy disappeared at the end of 3 months of treatment with Cogent db. There was also a significant reduction in polyphagia, polydipsia, and in the number of neuromuscular complaints, with a remarkable improvement in the well-being of the patients.

The main goal of this project was to carry out a preliminary study to evaluate the clinical usefulness of the herbal preparation (Cogent db) in patients with type 2 diabetes. Partly because of the limited experience with this drug, it was, in this trial, used as an adjuvant to standard allopathic antidiabetic agents. The results have been impressive in favor of Cogent db; however, the design of the study still leaves room for speculation as to whether the effect of Cogent db was additive or synergistic to those of the allopathic drugs. The lack of placebo preparation for the study has also introduced a confounding factor in the interpretation of the data that are nonetheless interesting. This poses a challenge for further studies. The current results, however, are consistent with animal studies, which show that Cogent db possesses a useful and unique antidiabetic profile. Our results are also consistent with the data (unpublished data) from a more recent double-blind placebo controlled study<sup>#</sup> in which Cogent db used as a main drug exhibited an antidiabetic profile. The results of this study show that Cogent db (as an adjuvant therapy):

1. Is a potent antidiabetic agent as revealed by its blood and urinary glucose lowering effect, significant reduction in Hb A<sub>1C</sub>, glycated hemoglobin, and proteinuria;
2. Within the period (3 months) of this study it does not cause hepatotoxicity, nephrotoxicity, or hematologic toxicity, thereby suggesting that it is safe within the limits of the tested parameters;
3. Has an ability to reduce the cholesterol and triglycerides, which are important risk factors for diabetes;
4. Restores liver enzymes to normal levels;

5. To a certain extent significantly normalizes the fasting insulin (experimental evidence in rodents showed that there was a rejuvenation of the pancreas after Cogent db treatment [Joy and Kuttan, 1998]. Whether this is true in human subjects needs to be evaluated).

The results of this study strongly support the anecdotal evidence it set out to assess. A double-blind placebo controlled study is proposed to further confirm/clarify the findings of this preliminary study.

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## REFERENCES

- John PC. Historical background. Textbook of Diabetes, 2nd ed., vol. I. London: Blackwell Science, 1997:101.
- Joy KL, Kuttan R. Anti-diabetic activity of Cogent db-a herbal preparation. *Amala Res Bull* 1998;18:109-114.
- McCarty D, Zimmet P. Diabetes—1994 to 2010: Global Estimate and Projections. International Diabetic Institute, Geneva: WHO, 2000.
- Mustaffa BE. Diabetes mellitus in Asia: An increasing health care burden. *Med Dig* 1998;16:5-9.
- Mustaffa BE. Strategies and Action Plans to improve diabetes care: The Malaysian model. *Journal of the American Medical Association Southeast Asia Nov/Dec* 1999;15(1):49-50.

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<sup>#</sup>Jayakumar RV. Clinical evaluation of herbal preparation in non insulin dependent diabetes mellitus. Government of Kerala State Board of Medical Research. Final Report. April 1999–September 2000.